

ESE 4020 / 5420 — Midterm 1

Modules 1 and 2. Every formula below is used by at least one practice question.

Derivations in grey. Appendix 2F is not examined.

1. Probability, conditioning, Bayes

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

and $\Pr(A \cup B) = \Pr(A) + \Pr(B)$ when A and B are disjoint, since then $\Pr(A \cap B) = 0$.

$$\Pr(A | B) = \frac{\Pr(A \cap B)}{\Pr(B)} \quad \text{for } \Pr(B) > 0, \quad \text{so} \quad \Pr(A \cap B) = \Pr(A | B) \Pr(B).$$

A and B are independent when $\Pr(A | B) = \Pr(A)$, equivalently $\Pr(A \cap B) = \Pr(A)\Pr(B)$.

For a partition $B_1, \dots, B_k$ of the sample space,

$$\Pr(A) = \sum_i \Pr(A | B_i) \Pr(B_i) = \sum_i \Pr(A \cap B_i).$$

$$\Pr(B | A) = \frac{\Pr(A | B) \Pr(B)}{\Pr(A)} = \frac{\Pr(A | B) \Pr(B)}{\sum_i \Pr(A | B_i) \Pr(B_i)}$$

Write $\Pr(A \cap B)$ two ways, $\Pr(A | B)\Pr(B) = \Pr(B | A)\Pr(A)$, divide by $\Pr(A)$, then expand $\Pr(A)$ by total probability. A small $\Pr(B)$ keeps the answer small even when the test is accurate.

2. One random variable

$$p_X(x) = \Pr(X = x) \quad F_X(x) = \Pr(X \leq x) \quad f_X = dF_X/dx.$$

A valid density satisfies all three of: $f_X(x) \geq 0$ everywhere; total area $\int f_X = 1$; and $\Pr(X \in A) = \int_A f_X(x) dx$. Being bounded by 1 or nondecreasing belongs to the CDF, not the density. For continuous X , $\Pr(X = a) = 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x) \quad \text{or} \quad \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Expectation is linear: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ and $\mathbb{E}[f + g] = \mathbb{E}[f] + \mathbb{E}[g]$.

$$\text{Var}[X] = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[X^2] - \mu_X^2$$

Expand the square: $\mathbb{E}[X^2] - 2\mu_X\mathbb{E}[X] + \mu_X^2 = \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2$.

Four facts that hold for every X and constants a, c :

$$\text{Var}[X] \geq 0, \quad \text{Var}[X + c] = \text{Var}[X], \quad \text{Var}[aX] = a^2 \text{Var}[X],$$

and $\sigma_X = \sqrt{\text{Var}[X]}$ carries the same units as X . Shifting moves the distribution without spreading it; scaling by a scales distances by a and squared distances by a^2 . For independent X, Y , $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$.

$$\text{Uniform on } [a, b]: \quad \mu = \frac{a+b}{2}, \quad \text{Var} = \frac{(b-a)^2}{12}, \quad \sigma = \frac{b-a}{\sqrt{12}}. \quad \int_a^b \frac{(x-\mu)^2}{b-a} dx = \frac{1}{b-a} \left[\frac{(x-\mu)^3}{3} \right]_a^b = \frac{(b-a)^2}{12}.$$

Only the width matters, not where the interval sits.

For $X \sim \mathcal{N}(\mu, \sigma^2)$, the 68–95–99.7 rule: about 68% of the mass lies in $[\mu - \sigma, \mu + \sigma]$, about 95% in $[\mu - 2\sigma, \mu + 2\sigma]$, about 99.7% in $[\mu - 3\sigma, \mu + 3\sigma]$. Worth memorising — it reads off Gaussian probabilities with no table and no calculator. The density formula itself is not needed.

3. Two random variables

Marginals: $p_X(x) = \sum_y p_{XY}(x, y)$, $f_X(x) = \int f_{XY}(x, y) dy$ — sum or integrate the other variable away.

Conditionals: $p_{Y|X}(y | x) = \frac{p_{XY}(x, y)}{p_X(x)}$, $f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)}$.

Independence: $p_{XY}(x, y) = p_X(x)p_Y(y)$ for all x, y (likewise for densities).

$$\text{Cov}[X, Y] = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y$$

Expand the product and use linearity; two of the three cross terms collapse to $-\mu_X\mu_Y$.

$\text{Cov}[X, X] = \text{Var}[X]$. Correlation $\rho = \frac{\text{Cov}[X, Y]}{\sigma_X\sigma_Y} \in [-1, 1]$. Independence implies $\text{Cov} = 0$, but *not* the reverse: covariance sees only linear association. Take X symmetric about zero and $Y = X^2$ — the covariance vanishes though Y is a function of X .

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y]$$

Square $(X - \mu_X) + (Y - \mu_Y)$ and take expectations term by term. The covariance enters twice.

Conditional expectation, $\mathbb{E}[Y | X = x] = \sum_y y p_{Y|X}(y | x)$ or $\int y f_{Y|X}(y | x) dy$, with four properties worth knowing:

- $\mathbb{E}[aY + b | X] = a \mathbb{E}[Y | X] + b$ — it is linear, like ordinary expectation.
- $\mathbb{E}[Y | X = x]$ is a *number* once x is fixed; $\mathbb{E}[Y | X]$ is a *random variable*, being a function of X .
- If X and Y are independent, $\mathbb{E}[Y | X] = \mathbb{E}[Y]$ — the conditioning tells you nothing.
- If $Y = g(X)$, then $\mathbb{E}[Y | X] = g(X)$ — knowing X determines Y exactly.

$\text{Var}[Y | X = x]$ measures how much uncertainty about Y is left once $X = x$ is known.

4. Law of large numbers, CLT, sampling distribution

For i.i.d. $X_1, \dots, X_N$ with mean μ and standard deviation σ , write $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$. Then

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}[\bar{X}] = \frac{\sigma^2}{N}, \quad \text{sd}[\bar{X}] = \frac{\sigma}{\sqrt{N}}.$$

Linearity gives the mean. For the variance, $\text{Var}[\frac{1}{N} \sum X_i] = \frac{1}{N^2} \sum \text{Var}[X_i] = \frac{N\sigma^2}{N^2}$, using independence to drop the covariances. Because of the square root, quadrupling N halves the standard deviation — four times the data for twice the precision.

Central limit theorem: $Z_N = \frac{\sum_i X_i - N\mu}{\sigma\sqrt{N}} \rightarrow \mathcal{N}(0, 1)$, so $\bar{X} \approx \mathcal{N}(\mu, \sigma^2/N)$ for large N , whatever the shape of the data.

The *sampling distribution* is the distribution of the estimator across repeated datasets, not the histogram of one dataset. Your single $\bar{X}$ is one draw from it, and it narrows as N grows.

5. The statistical learning problem

N is the number of samples, p the number of input features. Regression has a continuous output; classification has a discrete label.

Additive error model: $Y = f(\vec{X}) + \varepsilon$, with $\mathbb{E}[\varepsilon] = 0$, $\text{Var}[\varepsilon] = \sigma_\varepsilon^2$, and ε independent of $\vec{X}$. The regression function f is unknown and must be estimated. The conditional density is the noise density shifted by f :

$$f_{Y|X}(y | x) = f_\varepsilon(y - f(x))$$

$$\mathbb{E}[Y | X = x] = f(x)$$

$$\mathbb{E}[f(X) + \varepsilon | X = x] = f(x) + \mathbb{E}[\varepsilon] = f(x).$$

Under squared-error loss the best predictor at x_0 is the conditional mean $\mathbb{E}[Y | X = x_0]$; under absolute error it would be the conditional median. This is what makes the conditional mean the target of regression, and it is why averaging the y_i near x estimates $f(x)$: the conditional sample mean approaches the conditional expectation by the LLN, and the zero-mean noise averages away.

6. Local averaging, kNN, the curse of dimensionality

Ball of radius r about a query point: $B_r(x_0) = \{i : \|\vec{x}_i - \vec{x}_0\| \leq r\}$, and

$$\hat{f}_r(x_0) = \frac{1}{|B_r(x_0)|} \sum_{i \in B_r(x_0)} y_i.$$

kNN fixes the *count* k rather than the radius: $\hat{f}_{k\text{NN}}(x_0) = \frac{1}{k} \sum_{i \in D_k(x_0)} y_i$, where $D_k(x_0)$ holds the k nearest samples. For classification, take the most frequent label among them:

$$\text{Class}(x_0) = \arg \max_c \sum_{i \in D_k(x_0)} \mathbf{1}\{y_i = c\}.$$

k is a hyperparameter, chosen not fitted. As k grows the fit gets smoother, the spread across datasets falls and the bias typically rises; at $k = N$ the prediction is the global average. At $k = 1$ the boundary is ragged and the model chases noise.

For N points spread uniformly over the unit ball in p dimensions,

$$\mathbb{E}[|B_r(x_0)|] = Nr^p.$$

Expected count = $N \times (\text{ball volume} \div \text{region volume})$. Both volumes carry the same constant k_p , so it cancels and only r^p survives. With $r < 1$, each extra dimension multiplies the count by another factor of r .

So in high dimensions a small ball is almost always empty: local averaging needs exponentially more data as p grows, and to keep the count fixed r must approach 1, at which point the neighbourhood is no longer local. This is the curse of dimensionality — a fact about geometry, not about noise.

7. Parametric models and training

Instead of averaging neighbours, assume a shape with m tunable parameters θ , so that every sample informs every parameter:

$$\hat{f}(\vec{x}_i; \theta) = \theta_0 + \sum_{j=1}^p \theta_j x_{i,j} \quad \text{or} \quad \hat{f}(x_i; \theta) = \sum_{k=0}^d \theta_k x_i^k.$$

$$\text{MSE}(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}(\vec{x}_i; \theta))^2, \quad \hat{\theta} = \arg \min_{\theta} \text{MSE}(\theta)$$

Gradient descent steps *against* the gradient, which points uphill:

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla \text{MSE}(\theta^{(t)}), \quad \alpha > 0.$$

A learning rate α that is too large oscillates or diverges; too small and convergence crawls ($\alpha = 0$ makes no progress at all). α is chosen by us, never fitted by the algorithm.

Stochastic gradient descent uses a random mini-batch $M \subset D$ each step: cheaper per step, and the noise in the gradient can carry the path out of a shallow basin. Momentum keeps a running average of recent gradients, damping oscillation across a narrow valley without storing every past gradient. Adaptive methods use a different effective step size in each coordinate.

Hyperparameters (k, d, H, α) are chosen by us; parameters (θ) are fitted to the data. Residuals and noise realisations are neither.

8. Flexibility and the bias–variance tradeoff

$$\text{MSE}_{\text{Tr}} = \frac{1}{N_{\text{Tr}}} \sum_{D_{\text{Tr}}} (y_i - \hat{f}(\vec{x}_i))^2, \quad \text{MSE}_{\text{Te}} = \frac{1}{N_{\text{Te}}} \sum_{D_{\text{Te}}} (y_i - \hat{f}(\vec{x}_i))^2$$

Within a nested family, more flexibility never increases the training error, so MSE_{Tr} falls monotonically and cannot tell you when to stop. The test error is U-shaped, and its minimum marks the right flexibility. The dataset is itself random, so a refit on a fresh dataset gives a different $\hat{f}$, and the prediction $\hat{f}(x_0)$ is a random variable. Averaging the predictions from many training sets gives $\bar{f}(x_0)$, and

$$\text{bias}(x_0) = \bar{f}(x_0) - f(x_0),$$

the gap between that average and the truth. The scatter of the individual predictions about their own average is the separate *spread* term. Then

$$\text{MSE}_{\text{Te}} \approx \underbrace{\text{bias}^2 + \text{spread}^2}_{\text{reducible}} + \underbrace{\sigma_{\varepsilon}^2}_{\text{irreducible}}$$

As flexibility rises the squared bias falls and the squared spread rises; σ_{ε}^2 depends on Nature, not on the model, and no amount of flexibility reduces it. It is a floor MSE_{Te} cannot go below, since a perfect $\hat{f} = f$ still leaves $\frac{1}{N_{\text{Te}}} \sum \varepsilon_i^2 \approx \sigma_{\varepsilon}^2$ — and in that case both bias and spread vanish.

Honest model selection: split D into $D_{\text{Tr}} / D_{\text{Va}} / D_{\text{Te}}$ (roughly 60/20/20). Fit candidates on D_{Tr} , choose the flexibility by MSE_{Va} , and touch D_{Te} once, at the very end. Choosing on the test set is data leakage: you would be reporting the best of many tries, which is optimistic.